

Autonomous sensor-data market

Pragma.Vision **DECISION REQUESTED — MILESTONE 1**

A device agent sells single sensor readings to any agent that asks, at \$0.001 per call, with no account, no API key and no prior contract. *The settlement path is already proven on Base Sepolia; what remains is the layer above it.*



THE PROBLEM

why this exchange does not happen today

An agent that needs one sensor reading has two options, and both are bad. It can hold an API key — which means a human signed a contract, opened an account and configured billing in advance, for a transaction worth a tenth of a cent. Or it goes without the data.

This is why machine-to-machine data markets have been predicted for twenty years and have not appeared: the smallest unit of commercial relationship has always been larger than the smallest unit of useful data. The three protocols above collapse that unit to a single HTTP request.

ACTORS

no prior relationship on either side

PARTY	IS	WANTS
Seller	Device twin owning one simulated environmental sensor	To sell readings to strangers with zero onboarding
Buyer	Monitoring or forecasting agent	One reading, now, without an account or a key
Facilitator	Public x402 service	Verifies the signature, settles on-chain, pays gas

UNIT ECONOMICS

price set in the PoC

Price per reading	\$0.001 USDC
Gas cost to seller	zero
Buyer ETH requirement	none — verified
Minimum transaction	one call
Agent polling 1/min	\$1.44 / day
Ten such buyers	≈ \$430 / month

Those last two rows are arithmetic from the PoC price, not measured demand. The defensible claim is smaller and more useful: **the marginal cost of accepting one more buyer is zero**, because there is no onboarding step to perform. What becomes sellable is data previously given away or locked behind an enterprise agreement — because there was no way to charge a stranger a tenth of a cent.

SIMULATED VERSUS REAL

stated plainly

SIMULATED	The sensor hardware. No physical device exists.
SIMULATED	The telemetry values, generated in the Worker.
ZERO VALUE	The money. Faucet USDC on Base Sepolia.
REAL	A2A discovery — a served Agent Card, fetched over HTTP.
REAL	The mandate — signed JSON. Schema is mine; no standard exists yet.
REAL	The 402 exchange, verified against a public facilitator.
REAL	The settlement — an on-chain transaction, independently checkable.

The claim under test is not *this device exists*. It is: **when a device does exist, this is the mechanism that lets it sell to strangers**. Every part of that mechanism is real; only the thing generating the numbers is fake.

DEMO STORYBOARD

what a stranger sees in 30 seconds

- The closed door.** A visitor calls /reading with no payment and gets 402 with machine-readable terms. A browser at the same URL renders a human-readable paywall instead. One endpoint, two audiences, no second integration.
- The transaction.** A live feed shows the buyer discovering the seller, checking its mandate, signing, paying, receiving. Each settlement is a row linking to the block explorer. The visitor clicks one and sees the transfer on-chain.
- The ledger.** Settlements today, total volume, distinct payers — with a permanent **TESTNET** badge beside them. The honesty is not a disclaimer; it is what makes the counters believable.

RISKS AND MITIGATIONS

known before the build starts

RISK	MITIGATION
Facilitator is a single point of failure	Fail closed — 503 with Retry-After. Never serve unpaid. URL stays configurable.
Payment proof replayed	Idempotency key on the payment signature in KV, 24h TTL, replays rejected loudly.
Autonomous buyer overspends	Hard budget cap and kill switch. An unbounded autonomous spender is a bug, not a feature.
Two incompatible x402 package generations	Pinned versions, documented in the README. This cost me time and will cost every newcomer time.
Nobody outside the project ever pays	Five-minute quickstart plus hand-onboarding the first three developers. If it still fails, that failure is itself a reportable finding.

OPEN QUESTIONS

for the Friday decision

- Q1** Should the demo device be generic, or aligned to a sector where Pragma.Vision already has an audience? The protocol work is identical; the story and the inbound traffic are not.
- Q2** Does soft.house have an endpoint that could plausibly be metered this way later? If so, the tutorial should be written against a shape that matches it.
- Q3** The mandate schema is mine to invent. Keep it minimal for demo legibility, or track the AP2 draft closely enough to be reusable if AP2 stabilises?
- Q4** For Week 7, is a *recent payers* board on the public demo page acceptable? It shows wallet addresses — public, but still identifiers.

WHAT IS ALREADY BUILT

as of 27 July

A Cloudflare Worker gates GET /reading with x402. An unpaid request returns 402 with requirements in a PAYMENT-REQUIRED header. A Node buyer agent decoded them, signed an EIP-3009 authorization, retried, and the facilitator settled on Base Sepolia. The Worker returned the reading plus a receipt. **No human, no card, no account**. Remaining for PoC v1: discovery, mandates, replay protection, demo surface.

Recommendation — build this. Pay-per-inference is planned as a second resource type behind the same gate in Week 5, which makes it an addition rather than an alternative. A third candidate, metered device-time rental, is the most commercially interesting of the three and is deliberately deferred: exclusive time-bounded leasing raises dispute and refund questions that need a mandate layer nobody has standardised yet. It is recorded here as the strongest argument for continued work rather than as a competing option for this sprint.

EVIDENCE — FIRST SETTLEMENT
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f7965470a27ae2b93f4d1f3
Base Sepolia · eip155:84532 · 0.001 USDC
Transcript: docs/402-transcript.txt
TESTNET ONLY — NO REAL VALUE